# TechBes Main App – Complete Technical Documentation

System: TechBes Customer Marketplace, Digital Services Portal & Resident Platform
Target Application: `main_app/` (Next.js 16.2.0, React 19.2.4, Tailwind CSS 4.2.0)
Backend Core: `BE/` (Node.js 18+, Express 4.21.2, Socket.io 4.8.3, MongoDB Atlas / Mongoose 8.9.3)
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## Document Control

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1.0.0 | 2026-10-19 | DeepMind Antigravity Audit | Comprehensive technical documentation covering 28 customer marketplace modules, dynamic routing, rewrites, AI assistant, AMC calculator, customer dashboard, and resident activation.
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## 1. Executive Summary

### 1.1 Purpose and Audit Scope

The TechBes Main App (`main_app/`) is the customer-facing digital marketplace, booking engine, quotation submission portal, and resident services platform for TechBes Services Private Limited. It caters to commercial enterprises, homeowners, housing society residents, and facility managers across the Greater Bangalore metropolitan region.

This technical documentation reflects the actual implemented architecture and codebase located at `c:\Work\technician_app\main_app` and its backend dependencies at `c:\Work\technician_app\BE`. The audit was conducted strictly in read-only mode to preserve customer records, payment integrity, and system stability.

### 1.2 Key Architectural Findings

- Next.js App Router Architecture: Built on Next.js 16.2.0 and React 19.2.4 with Tailwind CSS 4.2.0, utilizing both React Server Components (RSC) for SEO-optimized service pages and client-side interactive islands for booking wizards, audio recording, and map pickers.
- Transparent Server-Side API Rewriting: Rather than making direct cross-origin requests from the browser, `main_app/next.config.mjs` configures transparent reverse proxy rewrites:
- `/api/v2/:path*` $\rightarrow$ `${BACKEND_API_URL}/api/v2/:path*`

- `/api/payments/:path*` $\rightarrow$ `${BACKEND_API_URL}/api/payments/:path*`
- `/api/bookings/:path*` $\rightarrow$ `${BACKEND_API_URL}/api/bookings/:path*`
- `/leads/:path*` $\rightarrow$ `${BACKEND_API_URL}/leads/:path*`
- `/api/auth/:path*` $\rightarrow$ `${BACKEND_API_URL}/api/auth/:path*`

This prevents CORS issues in production, hides backend origin topography, and facilitates cookie-based session forwarding.

- Embedded AI Customer Assistant: Features an intelligent interactive widget (`AIAssistantWidget.tsx`) connected to Google Gemini 1.5 Pro via `/api/v2/ai/chat` and `/api/v2/ai/handoff`. The engine extracts camera requirements, calculates live pricing estimates, and converts user chat sessions into formal support tickets.
- Hyperlocal SEO & Answer Engine Optimization (AEO): Embedded datasets (`geo-data.ts` - 60 KB, `aeo-data.ts` - 23 KB) provide programmatic SEO metadata, JSON-LD structured schemas, and geographic locality landing configurations for Bangalore pin codes and areas.
- Dedicated Society Resident Activation: Implements a resident onboarding portal at `/resident/activate` enabling apartment society tenants and owners to claim flat profiles using 6-digit invitation codes.

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# 2. Project Scope and System Architecture

## 2.1 Scope Boundaries

The Main App encompasses the following customer-facing subsystems:

1. Digital Marketplace & Services Catalog: Dynamic browsing of CCTV installation, repair, networking, biometric, and security systems with customizable packages.
2. Commercial Quotation Engine: Voice note audio recording, image attachments, floor plan uploads, and interactive equipment requirement forms.
3. Checkout & Advance Payment: Real-time slot booking, address selection, coupon application, and Razorpay advance payment gateway.
4. Customer Operations Dashboard: 12-section portal for tracking active jobs, downloading signed digital worksheets, viewing quotation revisions, and managing saved addresses.
5. Annual Maintenance Contracts (AMC): Silver, Gold, and Diamond tier pricing calculator and instant contract purchase inquiry.
6. Society Resident Portal: Flat activation, maintenance complaint ticketing, and society notice boards.

## 2.2 System Architecture Diagram

The following Mermaid architecture diagram illustrates the component interactions across the Main App, Next.js edge rewrites, Backend API, and external service providers:

```
flowchart TB
    subgraph Client_Browser ["Customer Client Browser"]
        Marketplace_UI["Marketplace & Service Pages\n(React 19 / Tailwind 4)"]
        Booking_Wizard["Checkout & Booking Engine\n(cctv-checkout-view.tsx)"]
        AI_Widget["AI Assistant Chat Widget\n(AIAssistantWidget.tsx)"]
        Customer_Dash["Customer Dashboard\n(/dashboard/:path*)"]
        Resident_Portal["Resident Activation\n(/resident/activate)"]
    end
    subgraph Next_Edge ["Next.js App Server (main_app - :3000)"]
        Edge_Middleware["Proxy Middleware\n(proxy.ts - Route Guard)"]
        Next_Rewrites["Next.js Rewrites Engine\n(next.config.mjs)"]
        Dynamic_SSR["SSR & Metadata Engine\n(generateMetadata & JSON-LD)"]
    end
    subgraph Backend_Core ["TechBes Express Backend (BE - :5000)"]
        BE_Router["V2 API Gateway\n(/api/v2/*)"]
        AI_Service["AI Conversation Engine\n(Google Gemini Integration)"]
        Dispatch_Core["Booking & Dispatch Service\n(autoAssignTechnician)"]
        Quote_Service["Quote Management Engine\n(PDFKit Document Synthesis)"]
        Apartment_Core["Apartment & Resident Service\n(Ticket Routing)"]
    end
    subgraph External_Cloud ["Cloud Services"]
        Razorpay_Gate["Razorpay Payment Gateway"]
        OSM_Tiles["OpenStreetMap / Nominatim API"]
        Cloudinary_Store["Cloudinary Storage (Audio/Images)"]
        Gemini_Cloud["Google AI Studio (Gemini 1.5)"]
    end
    Marketplace_UI --> Edge_Middleware
    Customer_Dash --> Edge_Middleware
    Edge_Middleware --> Next_Rewrites
    Booking_Wizard --> Next_Rewrites
    AI_Widget --> Next_Rewrites
    Resident_Portal --> Next_Rewrites
    Next_Rewrites -- "/api/v2/*" --> BE_Router
    Next_Rewrites -- "/api/geocode" --> OSM_Tiles
    BE_Router --> AI_Service --> Gemini_Cloud
    BE_Router --> Dispatch_Core
    BE_Router --> Quote_Service
    BE_Router --> Apartment_Core
    Booking_Wizard --> Razorpay_Gate
    Booking_Wizard --> Cloudinary_Store
```

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# 3. Technology Stack and Verified Dependencies

### 3.1 Main App Stack (`main_app/package.json`)

The dependencies and runtime versions verified directly from `main_app/package.json` are:

Category | Package Name | Verified Version | Purpose in Main App
Framework | `next` | `16.2.0` | React Server Components, App Router, Image Optimization, Reverse Proxy Rewrites
Runtime Library | `react` / `react-dom` | `19.2.4` | Component lifecycle, concurrent rendering, and DOM reconciliation
Styling & Design | `tailwindcss` | `^4.2.0` | Next-generation utility-first styling system
PostCSS Pipeline | `@tailwindcss/postcss` | `^4.2.0` | PostCSS plugin for Tailwind CSS v4
UI Primitives | `@radix-ui/react-*` | Various | 22 accessible headless component primitives (Dialog, Tabs, Accordion, Select, Tooltip, etc.)
Icons Library | `lucide-react` | `^0.564.0` | Vector UI icons for marketplace categories and navigation
Animations | `framer-motion` | `^12.42.2` | Fluid micro-interactions, modal transitions, and banner animations
Form Handling | `react-hook-form` | `^7.54.1` | Performant, uncontrolled form state management
Form Validation | `@hookform/resolvers` | `^3.9.1` | Zod schema validation bridge for React Hook Form
Data Validation | `zod` | `^3.24.1` | TypeScript-first schema declaration and input sanitization
Mapping & Geospatial | `leaflet` / `react-leaflet` | `^1.9.4` / `^5.0.0` | Interactive map picker for customer address pin drops
Map TypeScript Types | `@types/leaflet` | `^1.9.21` | Typings for Leaflet map instances and layer controls
Carousel & Slider | `embla-carousel-react` | `8.6.0` | Touch-enabled responsive carousels for hero banners
Toasts & Alerts | `sonner` | `^1.7.1` | Toast notification manager
Drawer Primitive | `vaul` | `^1.1.2` | Mobile-friendly swipeable drawer component
OTP Verification | `input-otp` | `1.4.2` | One-Time Password input box for phone verification
Date Operations | `date-fns` | `4.1.0` | Date formatting and slot calculation utilities
Date Picker | `react-day-picker` | `9.13.2` | Interactive calendar date selector for booking visits
Class Utilities | `clsx` / `tailwind-merge` | `^2.1.1` / `^3.3.1` | Dynamic conditional class merging (`cn()` utility)
Analytics Engine | `@vercel/analytics` | `1.6.1` | Web vitals and client-side performance monitoring

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## 4. Repository and Folder Structure

### 4.1 Main App Layout (`main_app/`)

```
c:\Work\technician_app\main_app\
% % %  .env.example                    # Reference environment configuration
% % %  .env                            # Active local environment variables
% % %  next.config.mjs                 # Next.js config (Rewrites, Security Headers, Redirects)
% % %  package.json                    # Dependencies, scripts, and engine constraints
% % %  tsconfig.json                   # TypeScript configuration (paths: @/* -> ./*)
% % %  proxy.ts                        # Next.js middleware guarding /dashboard and /login
% % %  public/                         # Static images, hero illustration, brand logos, manifest
% % %  app/                            # Next.js App Router root
%    % % %  layout.tsx                  # Root HTML shell with Navbar, Footer, and AIAssistant
%    % % %  page.tsx                    # Homepage with Hero, Categories, Trust, and Services
%    % % %  globals.css                 # Tailwind CSS v4 theme variables and global styles
%    % % %  sitemap.ts                  # Dynamic programmatic XML sitemap generator
%    % % %  robots.ts                   # Dynamic robots.txt generator
%    % % %  not-found.tsx               # Branded 404 error page with search redirect
%    % % %  loading.tsx                 # Global loading skeleton
%    % % %  services/                   # Service catalog directory
%    %    % % %  page.tsx               # Service catalog listing page
%    %    % % %  [slug]/page.tsx        # Dynamic service details page with 301 redirects
%    % % %  amc/page.tsx                # AMC plans, interactive calculator, and inquiry form
%    % % %  get-a-quote/page.tsx        # Commercial quotation submission portal
%    % % %  quote/                      # Quotation review and conversion
%    %    % % %  page.tsx               # Quote lookup by ID
%    %    % % %  success/page.tsx       # Quote submission confirmation
%    % % %  cart/page.tsx               # Shopping cart view for CCTV packages and hardware
%    % % %  checkout/page.tsx           # Multi-step checkout, slot booking & advance payment
%    % % %  booking-success/page.tsx    # Booking confirmation and order receipt
%    % % %  resident/                   # Apartment society resident portal
%    %    % % %  activate/page.tsx      # 6-digit code flat account activation
%    % % %  dashboard/                  # Authenticated customer operations portal
%    %    % % %  page.tsx               # Dashboard overview & active orders
%    %    % % %  bookings/              # Booking history & live technician tracking
%    %    % % %  quotes/                # Quotation status & PDF download
%    %    % % %  amc/                   # Active AMC contracts & visit progress
%    %    % % %  service-report/        # Digital job completion worksheets
%    %    % % %  addresses/             # Saved addresses & pin drops
%    %    % % %  payments/              # Payment transaction ledger
%    %    % % %  invoices/              # Downloadable tax invoices
%    %    % % %  reviews/               # Service feedback & rating submission
%    %    % % %  wallet/                # Credits and loyalty points
%    %    % % %  support/               # Helpdesk support tickets
%    %    % % %  notifications/         # Customer alerts feed
%    %    % % %  profile/               # Contact details & password security
%    % % %  admin/payments/page.tsx     # Internal payment monitoring stub
%    % % %  api/                        # Local API routes (geocode, cart, auth)
%    % % %  (compliance)/               # Legal and policy pages (privacy, terms, etc.)
% % %  components/                     # UI and layout component hierarchy
%    % % %  AIAssistantWidget.tsx       # Floating Google Gemini chatbot widget
%    % % %  navbar.tsx                  # Responsive header with location selector & cart badge
%    % % %  footer.tsx                  # Comprehensive footer with SEO links and contact info
%    % % %  hero-section.tsx            # Dynamic hero banner with search & CTA
%    % % %  category-section.tsx        # Service category cards grid
```

```
    %    % % %  popular-services.tsx          # High-demand services showcase
    %    % % %  trust-banner.tsx              # Guarantees, warranties, and certifications banner
    %    % % %  booking/                      # Booking modal and LocationPicker (Leaflet)
    %    % % %  cctv/                         # CCTV package customizer and QuoteLocationPicker
    %    % % %  checkout/                     # cctv-checkout-view.tsx (Razorpay integration)
    %    % % %  quotation/                    # ProductAutocomplete and VoiceNoteRecorder
    %    % % %  services/                     # service-detail-view.tsx and package selectors
    %    % % %  ui/                           # 30+ reusable Shadcn UI components
    % % %  core/                            # Architecture core (API client, Auth session)
    %    % % %  api/                          # api-client.ts, backend-fetch.ts, config.ts
    %    % % %  auth/                         # session.ts (cookie helpers, auth keys)
    % % %  lib/                             # Data models, helpers, and static datasets
         % % %  aeo-data.ts                  # Answer Engine Optimization semantic dataset
         % % %  geo-data.ts                  # Bangalore localities and hyperlocal landing data
         % % %  cctv-api.ts                  # CCTV pricing calculation and hardware models
         % % %  catalog-api.ts               # Service catalog API client
         % % %  cctv-cart.ts                 # Cart state persistence and total calculations
         % % %  marketplace-data.ts          # Marketplace service definitions and fallback catalog
         % % %  seo-helpers.ts               # OpenGraph, Twitter card, and JSON-LD schema builder
```

---

## 5. Complete Module and Submodule Inventory

The Main App codebase implements 28 customer-facing functional modules. Each is documented below with its business purpose, entry route, key components, RBAC restrictions, APIs used, and underlying database models.

**# | Module Name | Business Purpose | Frontend Route | Key Component(s) | Access Role | API Endpoints Invoked | Primary DB Models | Status**

1 | Marketplace Home | Landing page with dynamic search, category browsing, and trust badges | `/` | `page.tsx`, `hero-section.tsx`, `category-section.tsx`, `popular-services.tsx` | Public | `/api/v2/catalog/categories` | `Category`, `SubCategory` | Fully Implemented

2 | Service Details & Packages | View service specifications, pricing starts from, included/excluded items, and booking questions | `/services/[slug]` | `services/[slug]/page.tsx`, `service-detail-view.tsx` | Public | `/api/v2/catalog/subcategories/:slug` | `SubCategory`, `Category` | Fully Implemented

3 | Commercial Quotation Wizard | Submit commercial security requirements with voice note recording, images, and map pin | `/get-a-quote` | `get-a-quote/page.tsx`, `VoiceNoteRecorder.tsx`, `QuoteLocationPicker.tsx` | Public / Customer | `POST /api/v2/quotes`, `POST /api/v2/quotes/voice-note` | `QuoteRequest`, `Material` | Fully Implemented

4 | Quotation Review & Conversion | Inspect priced quotation, line items, 18% GST breakdown, download PDF, and pay advance to convert | `/quote` | `quote/page.tsx`, `quote/success/page.tsx` | Public / Customer | `GET /api/v2/quotes/:id`, `GET /.../pdf`, `POST /.../create-payment-order` | `QuoteRequest`, `Job`, `Payment` | Fully Implemented

5 | AMC Pricing & Calculator | Calculate annual maintenance contract fees for CCTV systems based on camera counts; submit inquiry | `/amc` | `amc/page.tsx` | Public / Customer | `POST /api/v2/quotes` | `AmcContract`, `QuoteRequest` | Fully Implemented

6 | AI Security Assistant | Conversational Google Gemini chatbot providing CCTV advice, pricing estimates, and ticket creation | Global Widget | `AIAssistantWidget.tsx` | Public / Customer | `POST /api/v2/ai/chat`, `POST /api/v2/ai/handoff` | `SupportTicket`, `User` | Fully Implemented

7 | Interactive Checkout | Multi-step booking checkout: date/time slot selection, saved addresses, coupons, and Razorpay advance | `/checkout` | `checkout/page.tsx`, `cctv-checkout-view.tsx` | Customer (Auth required) | `POST /api/v2/bookings`, `POST /api/v2/payment/order`, `POST /.../verify` | `Job`, `Payment`, `Address` | Fully Implemented

8 | Digital Shopping Cart | View and adjust selected CCTV packages, hardware addons, camera types, and compute subtotal | `/cart` | `cart/page.tsx`, `cctv-cart.ts` | Public / Customer | `/api/cart` (local session or API bridge) | `Cart`, `Material` | Fully Implemented

9 | Booking Confirmation | Booking receipt with booking number, scheduled date, slot, and link to customer dashboard | `/booking-success` | `booking-success/page.tsx` | Customer | `GET /api/v2/bookings/:id` | `Job` | Fully Implemented

10 | Society Resident Activation | Allow apartment society tenants and owners to activate their flat resident portal using 6-digit code | `/resident/activate` | `resident/activate/page.tsx` | Public / Resident | `POST /api/v2/apartments/residents/activate` | `Apartment`, `User` | Fully Implemented

11 | Customer Dashboard Overview | Central customer hub displaying active bookings, pending quotes, AMC contracts, and quick actions | `/dashboard` | `dashboard/page.tsx` | Customer (Auth required) | `GET /api/dashboard`, `GET /api/v2/user/profile` | `User`, `Job`, `QuoteRequest` | Fully Implemented

12 | Customer Bookings Hub | View past and upcoming service visits, status progression (Assigned, Travelling, Started, Completed) | `/dashboard/bookings` | `dashboard/bookings/page.tsx` | Customer | `GET /api/v2/bookings` | `Job`, `JobStatusHistory` | Fully Implemented

13 | Customer Quotations Hub | View submitted quotation requests, review admin-priced estimates, download PDFs, and accept | `/dashboard/quotes` | `dashboard/quotes/page.tsx` | Customer | `GET /api/v2/quotes/my-quotes`, `GET /api/v2/quotes/:id/pdf` | `QuoteRequest` | Fully Implemented

14 | Customer AMC Contracts | View active Annual Maintenance Contracts, completed quarterly visits, and remaining service quota | `/dashboard/amc` | `dashboard/amc/page.tsx` | Customer | `GET /api/v2/amc/contracts` | `AmcContract` | Fully Implemented

15 | Customer Service Reports | View and download digital service worksheets submitted by technicians with before/after photos | `/dashboard/service-report` | `dashboard/service-report/[jobId]/page.tsx` | Customer | `GET /api/worksheets/job/:id`, `GET /api/v2/worksheets/:id/pdf` | `ServiceWorksheet`, `Job` | Fully Implemented

16 | Customer Address Book | Add, edit, and set default service locations with Bangalore locality search and pin drops | `/dashboard/addresses` | `dashboard/addresses/page.tsx`, `LocationPicker.tsx` | Customer | `GET/POST/PUT/DELETE /api/v2/user/addresses` | `Address`, `User` | Fully Implemented

17 | Customer Payments Ledger | Transaction history of all online advance payments, final settlements, and downloadable receipts | `/dashboard/payments` | `dashboard/payments/page.tsx` | Customer | `GET /api/v2/payment/user/history` | `Payment`, `Job` | Fully Implemented

18 | Customer Invoices | Download official GST-compliant tax invoices for completed service bookings | `/dashboard/invoices` | `dashboard/invoices/page.tsx` | Customer | `GET /api/v2/payment/invoices` | `Payment`, `Job` | Fully Implemented

19 | Customer Feedback & Ratings | Submit 1-5 star ratings and reviews for field technicians upon job completion | `/dashboard/reviews` | `dashboard/reviews/page.tsx` | Customer | `POST /api/v2/reviews` | `Review`, `Job`, `User` | Fully Implemented

20 | Customer Wallet & Credits | View loyalty credits, promotional cashbacks, and referral bonuses | `/dashboard/wallet` | `dashboard/wallet/page.tsx` | Customer | `GET /api/v2/wallet` | `Wallet`, `WalletTransaction` | Fully Implemented

21 | Customer Helpdesk Support | Raise maintenance tickets, attach photos, and receive replies from TechBes operations staff | `/dashboard/support` | `dashboard/support/page.tsx` | Customer | `GET/POST /api/v2/apartments/tickets` | `SupportTicket` | Fully Implemented

22 | Customer Notifications | In-app alerts for technician arrival, OTP generation, quote updates, and invoice readiness | `/dashboard/notifications` | `dashboard/notifications/page.tsx` | Customer | `GET /api/v2/notifications`, `PUT /api/v2/notifications/:id/read` | `Notification` | Fully Implemented

23 | Customer Profile & Security | Update personal name, email, phone number, and change account password | `/dashboard/profile` | `dashboard/profile/page.tsx` | Customer | `GET/PUT /api/v2/user/profile`, `PUT /api/v2/user/password` | `User` | Fully Implemented

24 | Customer Authentication | Mobile OTP login, email/password login, registration, and session token persistence | `/login` | `login/page.tsx` | Public | `POST /api/auth/login`, `POST /api/v2/otp/send`, `POST /.../verify` | `User`, `OtpVerification` | Fully Implemented

25 | Password Reset | Token-based password recovery via email link | `/reset-password` | `reset-password/page.tsx` | Public | `POST /api/auth/reset-password` | `User` | Fully Implemented
26 | Geocoding & Location Search | Reverse geocoding and autocomplete for Bangalore addresses using OpenStreetMap | API Route | `app/api/geocode/route.ts` | Public / Customer | `GET /api/geocode?action=search`, `GET /api/geocode?action=reverse` | Free Nominatim API | Fully Implemented
27 | XML Sitemap & SEO Engine | Automated XML sitemap generation indexing all dynamic services, localities, and categories | `/sitemap.xml` | `app/sitemap.ts`, `geo-data.ts`, `aeo-data.ts` | Search Engines | Dynamic XML response | Static Datasets & Catalog | Fully Implemented
28 | Admin Payment Monitor (Stub) | Internal payment monitoring console embedded inside the Main App structure | `/admin/payments` | `app/admin/payments/page.tsx` | Admin | `GET /api/v2/payment/admin/list`, `POST /.../refund`, `POST /.../retry` | `Payment` | Partially Implemented

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# 6. Frontend Pages, Routing, and Component Mapping

## 6.1 Route Redirection and SEO Mapping

`main_app/next.config.mjs` configures permanent 301 redirects to eliminate duplicate content and enforce canonical URL hierarchy:

- `www.techbes.co.in/:path*` $\rightarrow$ `https://techbes.co.in/:path*`
- `/services/cctv-installation` $\rightarrow$ `/services/install-new-cctv`
- `/services/cctv-repair` $\rightarrow$ `/services/repair-existing-cctv`
- `/services/cctv-maintenance` or `/services/cctv-amc` $\rightarrow$ `/services/maintenance-amc`

## 6.2 Customer Route Map Table

Frontend Route Path | Component File Location | Route Protection / Middleware | Data Dependencies | Primary User Action
`/` | `app/page.tsx` | Public | Category catalog, popular services | Browse services, search, open AI bot
`/services` | `app/services/page.tsx` | Public | Full service catalog | Filter by category, view pricing
`/services/[slug]` | `app/services/[slug]/page.tsx` | Public | Dynamic subcategory detail, AEO data | Read specs, configure package, book
`/amc` | `app/amc/page.tsx` | Public | Static pricing tiers, quote API | Calculate AMC cost, submit inquiry
`/get-a-quote` | `app/get-a-quote/page.tsx` | Public | Voice recorder, LocationPicker | Upload voice note, request custom quote
`/quote` | `app/quote/page.tsx` | Public / Customer | Quote API | Review quote, download PDF, pay
`/cart` | `app/cart/page.tsx` | Public / Customer | Local storage / cart state | Adjust item quantities, proceed
`/checkout` | `app/checkout/page.tsx` | Auth Required (`proxy.ts`) | User addresses, slot availability, Razorpay | Pick slot, select address, pay advance
`/booking-success` | `app/booking-success/page.tsx` | Customer | Booking record | View confirmation, track technician
`/resident/activate` | `app/resident/activate/page.tsx` | Public | Apartment resident activation API | Enter phone + 6-digit code, set password
`/dashboard` | `app/dashboard/page.tsx` | Auth Required (`proxy.ts`) | User profile, recent bookings | Overview of account activity
`/dashboard/bookings` | `app/dashboard/bookings/page.tsx` | Auth Required | User job records | Track technician arrival, inspect status
`/dashboard/quotes` | `app/dashboard/quotes/page.tsx` | Auth Required | User quotation requests | Review estimates, download quote PDFs
`/dashboard/amc` | `app/dashboard/amc/page.tsx` | Auth Required | User AMC contracts | View completed visits, schedule engineer
`/dashboard/service-report/[jobId]` | `app/dashboard/service-report/[jobId]/page.tsx` | Auth Required | ServiceWorksheet record | Inspect photo proofs, download worksheet PDF
`/dashboard/addresses` | `app/dashboard/addresses/page.tsx` | Auth Required | User address list | Add new address, drop map pin
`/dashboard/payments` | `app/dashboard/payments/page.tsx` | Auth Required | Payment ledger | View paid amounts, Razorpay payment IDs
`/dashboard/support` | `app/dashboard/support/page.tsx` | Auth Required | Support tickets | Submit complaint, reply to messages
`/login` | `app/login/page.tsx` | Guest Only (redirects to `/dashboard` if logged in) | Auth API | Login via mobile OTP or password
`/reset-password` | `app/reset-password/page.tsx` | Public | Password reset API | Enter new password using reset token

---

# 7. Complete Backend API Reference (Main App Facing)

The Main App interacts with the backend through transparent Next.js rewrites. The table below catalogs all primary customer-facing API endpoints:

**Method | Actual Target Endpoint | Calling Main App Component | Purpose | Auth Required | Request Body / Parameters | Response Schema | Controller Location | DB Models Involved**
GET | `/api/v2/catalog/categories` | `category-section.tsx` | Fetch service categories for homepage grid | No | None | `{ success: true, data: Category[] }` | `BE/routes/v2/categoryRoutesV2.js` | `Category`
GET | `/api/v2/catalog/subcategories/:slug` | `app/services/[slug]/page.tsx` | Get subcategory details, packages, and booking questions | No | Slug parameter | `{ success: true, data: SubCategory }` | `BE/routes/v2/categoryRoutesV2.js` | `SubCategory`
POST | `/api/v2/quotes/voice-note` | `VoiceNoteRecorder.tsx` | Upload voice note audio file for quotation | No | Multipart form: `audio` (up to 15 MB) | `{ success: true, url, duration, filename }` | `BE/controllers/v2/quoteControllerV2.js` | `QuoteRequest`
POST | `/api/v2/quotes` | `get-a-quote/page.tsx`, `amc/page.tsx` | Submit quotation request with requirements and voice note | Optional | `{ fullName, mobile, email, address, locality, propertyType, cameraCount, voiceNote, images }` | `{ success: true, data: { requestId } }` | `BE/controllers/v2/quoteControllerV2.js` | `QuoteRequest`, `User`
GET | `/api/v2/quotes/my-quotes` | `dashboard/quotes/page.tsx` | List customer's submitted quotation requests | Yes | None | `{ success: true, data: QuoteRequest[] }` | `BE/controllers/v2/quoteControllerV2.js` | `QuoteRequest`
GET | `/api/v2/quotes/:id` | `app/quote/page.tsx` | Get quotation details, priced items, and GST | Optional | ID parameter | `{ success: true, data: QuoteRequest }` | `BE/controllers/v2/quoteControllerV2.js` | `QuoteRequest`
GET | `/api/v2/quotes/:id/pdf` | `app/quote/page.tsx`, `dashboard/quotes` | Stream compiled official quotation PDF buffer | Optional | ID parameter | Binary PDF (`application/pdf`) | `BE/controllers/v2/quoteControllerV2.js` | `QuoteRequest`
POST | `/api/v2/quotes/:id/create-payment-order` | `app/quote/page.tsx` | Create Razorpay order to pay quote advance | Yes | None | `{ success: true, order: { id, amount, currency } }` | `BE/controllers/v2/quoteControllerV2.js` | `QuoteRequest`, `Payment`
POST | `/api/v2/quotes/:id/verify-payment` | `app/quote/page.tsx` | Verify Razorpay payment and convert quote to booking | Yes | `{ razorpayOrderId, razorpayPaymentId, razorpaySignature }` | `{ success: true, bookingId: string }` | `BE/controllers/v2/quoteControllerV2.js` | `QuoteRequest`, `Job`, `Payment`
POST | `/api/v2/bookings` | `cctv-checkout-view.tsx` | Create field service booking with slot and address | Yes | `{ serviceId, scheduledDate, scheduledTime, addressId, cctvDetails, totalAmount }` | `{ success: true, booking: Job }` | `BE/controllers/v2/jobControllerV2.js` | `Job`, `User`, `Address`
GET | `/api/v2/bookings` | `dashboard/bookings/page.tsx` | List customer's past and active service bookings | Yes | None | `{ success: true, data: Job[] }` | `BE/controllers/v2/jobControllerV2.js` | `Job`

POST | `/api/v2/payment/order` | `cctv-checkout-view.tsx` | Create Razorpay order for booking advance payment | Yes | `{ bookingId, amount, currency }` | `{ success: true, orderId: string, amount: number }` | `BE/controllers/v2/paymentControllerV2.js` | `Job`, `Payment`
POST | `/api/v2/payment/verify` | `cctv-checkout-view.tsx` | Verify Razorpay payment signature for booking | Yes | `{ orderId, paymentId, signature }` | `{ success: true, payment: Payment }` | `BE/controllers/v2/paymentControllerV2.js` | `Payment`, `Job`
POST | `/api/v2/ai/chat` | `AIAssistantWidget.tsx` | Chat with Google Gemini security advisor bot | No | `{ message, history }` | `{ success: true, reply, estimatedPrice, state }` | `BE/services/aiService.js` | None
POST | `/api/v2/ai/handoff` | `AIAssistantWidget.tsx` | Convert AI chat inquiry into a support ticket | Yes | `{ chatSummary, category, requirements }` | `{ success: true, ticketId: string }` | `BE/services/aiService.js` | `SupportTicket`
POST | `/api/v2/apartments/residents/activate` | `app/resident/activate/page.tsx` | Activate flat resident account using 6-digit code | No | `{ phone, activationCode, password }` | `{ success: true, token, user, resident }` | `BE/controllers/v2/apartmentControllerV2.js` | `Apartment`, `User`
GET | `/api/v2/user/addresses` | `cctv-checkout-view.tsx`, `dashboard/addresses` | Fetch customer's saved service addresses | Yes | None | `{ success: true, data: Address[] }` | `BE/controllers/v2/userControllerV2.js` | `Address`
POST | `/api/v2/user/addresses` | `dashboard/addresses/page.tsx` | Save new service address with geo-coordinates | Yes | `{ addressLine1, area, city, pincode, latitude, longitude, isDefault }` | `{ success: true, data: Address }` | `BE/controllers/v2/userControllerV2.js` | `Address`
GET | `/api/v2/amc/contracts` | `dashboard/amc/page.tsx` | List customer's active AMC service contracts | Yes | None | `{ success: true, data: AmcContract[] }` | `BE/controllers/v2/amcControllerV2.js` | `AmcContract`
GET | `/api/worksheets/job/:id` | `dashboard/service-report/page.tsx` | Get completed job service worksheet with photo proofs | Yes | Job ID parameter | `{ success: true, data: ServiceWorksheet }` | `BE/routes/v2/worksheetRoutesV2.js` | `ServiceWorksheet`
POST | `/api/v2/reviews` | `dashboard/reviews/page.tsx` | Submit customer review and star rating for completed job | Yes | `{ jobId, rating, comment }` | `{ success: true, data: Review }` | `BE/controllers/reviewController.js` | `Review`, `Job`, `User`
GET | `/api/geocode` | `LocationPicker.tsx`, `QuoteLocationPicker.tsx` | Search localities or reverse geocode lat/lng | No | Query: `action` ('search' \| 'reverse'), `q`, `lat`, `lng` | Array of geocoded address items | `main_app/app/api/geocode/route.ts` | OpenStreetMap Nominatim
---

## 8. Database Schemas, Models, and Entity Relationships

The Main App reads and creates documents in 15 distinct MongoDB collections via the Backend services. The primary entities are:
- `User`: Customer identity, mobile number, role (`client`, `resident`), customerId (`CUS-YYYY-NNNNNN`), password hash.
- `Job`: Service booking record, scheduledDate, scheduledTime, cctvDetails, totalAmount, paymentStatus (`pending`, `advance_paid`, `paid`), advancePaymentId.
- `QuoteRequest`: Commercial quotation, requestId (`QT-YYYY-NNNNN`), itemized equipment list, voice note Cloudinary URL, PDF URL, 18% GST calculation.
- `Apartment`: Housing society registry containing embedded `residents` with `activationCode`, `activationExpires`, flat number, and resident type.
- `SupportTicket`: Customer and resident complaints, ticketId (`TKT-YYYY-NNNNN`), messages thread, and status.
- `ServiceWorksheet`: Completed job worksheet, before/during/after Cloudinary photos, customer signature, labour/material costs, PDF URL.
- `Address`: Saved customer service locations with geographic coordinates and Bangalore pin codes.
- `Payment`: Razorpay payment records, orderId, paymentId, amount, status (`SUCCESS`, `PENDING`, `FAILED`).
- `AmcContract`: Contract tier (`Silver`, `Gold`, `Diamond`), contractId (`TB-AMC-...`), quarterly visits array.
- `Category` & `SubCategory`: Catalog hierarchy defining dynamic service offerings, packages, pricing starts from, and booking questionnaire forms.

---

## 9. Core Business Workflows and Status Lifecycles

### 9.1 Self-Service CCTV Booking & Advance Payment Workflow

1. Catalog Exploration: Customer navigates to `/services/install-new-cctv` (or is redirected from `/services/cctv-installation`).
2. Package Selection: Selects camera count, camera type (Dome/Bullet/IP/PTZ), and optional hardware addons.
3. Cart Addition: Customer adds package to cart (`lib/cctv-cart.ts`) or clicks Book Now, transitioning to `/checkout`.
4. Checkout Configuration (`cctv-checkout-view.tsx`):
- Customer logs in or verifies mobile number.
- Selects service address (or drops a pin on Leaflet `LocationPicker.tsx`).
- Selects scheduled date and preferred time slot (e.g. `10:00 AM - 01:00 PM` or `ASAP`).
5. Order Generation: Frontend calls `POST /api/v2/payment/order`. Backend creates a Razorpay order for the booking advance (default 299 INR or customized advance percentage).
6. Payment Gateway Interaction: Razorpay checkout modal opens. Customer pays via UPI, Card, Net Banking, or Wallet.
7. Signature Verification & Job Creation:
- On payment success, frontend receives `razorpay_payment_id`, `razorpay_order_id`, `razorpay_signature` and posts to `POST /api/v2/payment/verify`.
- Backend verifies HMAC SHA-256 signature using `RAZORPAY_KEY_SECRET`.
- Upon verification, backend creates the `Job` record with status `assigned`, marks `advancePaid: true`, logs `PaymentAudit`, and triggers automated technician dispatch.
8. Confirmation: Customer is redirected to `/booking-success?id=BOOK-...`.

```
sequenceDiagram
    autonumber
    actor Customer as Marketplace Customer
    participant UI as cctv-checkout-view.tsx
    participant Proxy as Next.js Rewrites (:3000)
    participant BE as paymentControllerV2.js (:5000)
    participant RZP as Razorpay Gateway
    participant DB as MongoDB (Job & Payment)
    Customer->>UI: Select Package, Address & Time Slot
```

```
UI->>Proxy: POST /api/v2/payment/order
Proxy->>BE: Forward to /api/v2/payment/order
BE->>RZP: Create Order (Razorpay SDK)
RZP-->>BE: Return order_id
BE-->>UI: Return order_id & amount
UI->>RZP: Open Razorpay Checkout Modal
Customer->>RZP: Complete UPI / Card Payment
RZP-->>UI: Return payment_id & signature
UI->>Proxy: POST /api/v2/payment/verify
Proxy->>BE: Forward to /api/v2/payment/verify
BE->>BE: Verify HMAC SHA-256 Signature
BE->>DB: Create Job (advancePaid: true) & Record Payment
BE-->>UI: Success: { bookingId: "BOOK-..." }
UI-->>Customer: Redirect to /booking-success
```

### 9.2 Society Resident Portal Activation Workflow

1. Notice / SMS Delivery: Society resident receives an SMS or society email containing their flat details and a 6-digit activation code (generated by society admin in `ApartmentManagementPage.jsx`).
2. Accessing Portal: Resident navigates to `https://techbes.co.in/resident/activate?phone=9876543210&code=123456`.
3. Form Pre-fill & Credential Setup: Phone and code are automatically populated from query params. Resident enters and confirms a secure password.
4. Backend Verification:
- Frontend calls `POST /api/v2/apartments/residents/activate`.
- Backend locates the `Apartment` document containing a matching resident record with `activationCode` and verifies `activationExpires > Date.now()`.
- Creates or updates a `User` account with `role: 'resident'`, links `apartmentId`, `flatNumber`, and `block`.
- Marks resident record as `accountStatus: 'active'` and clears `activationCode`.
- Issues a JWT session token.
5. Session Initiation: Frontend stores the JWT in `localStorage` and `auth-token` cookie, redirecting the resident to their active resident dashboard.

---

## 10. Authentication, Authorization, and Session Security

### 10.1 Authentication Architecture

- Session Tokens: The Main App uses JSON Web Tokens (JWT) signed by the backend HMAC secret.
- Proxy Middleware Guard (`proxy.ts`):
- Next.js edge middleware intercepts all requests matching `/dashboard/:path*`.
- Checks for the presence of the `auth-token` cookie (key defined in `core/auth/session.ts`).
- If missing, redirects the user to `/login?redirect=/dashboard/...`.
- If an authenticated user navigates to `/login`, `proxy.ts` automatically redirects them to `/dashboard`.
- Client Storage: Upon login or activation, the token is saved to `localStorage.setItem('techbes_backend_token', token)` and `localStorage.setItem('auth-token', token)` to ensure client API fetchers can inject the `Authorization: Bearer <token>` header.

---

## 11. External Third-Party Integrations

### 11.1 Integration Registry

Provider | Purpose | Implementation Location | Credentials Key
Razorpay | Client advance payment collection and verification | `cctv-checkout-view.tsx`, `BE/config/razorpay.js` | `RAZORPAY_KEY_ID`, `RAZORPAY_KEY_SECRET`
Google Gemini AI | Conversational assistant widget | `AIAssistantWidget.tsx`, `BE/services/aiService.js` | `GEMINI_API_KEY`
OpenStreetMap / Nominatim | Reverse geocoding and Bangalore locality autocomplete | `main_app/app/api/geocode/route.ts` | Free Public API (with custom User-Agent)
Leaflet & OpenStreetMap | Interactive map pin drop for customer address selection | `components/booking/LocationPicker.tsx` | Tile Layer: `tile.openstreetmap.org`
Vercel Web Analytics | Client-side performance and web vitals tracking | `main_app/package.json` (`@vercel/analytics`) | Embedded Vercel environment
Cloudinary | Voice note audio and quotation image proof hosting | `VoiceNoteRecorder.tsx`, `BE/routes/v2/uploadRoutesV2.js` | `CLOUDINARY_CLOUD_NAME`, `CLOUDINARY_API_KEY`

---

## 12. Environment Configuration Reference

The following environment variables were verified from `main_app/.env.example` and `main_app/core/api/config.ts`:

Variable Name | Required | Default / Expected Format | Functional Purpose
`BACKEND_API_URL` | Yes | `https://api.techbes.co.in` or `http://localhost:5000` | Target URL used by `next.config.mjs` for server-side API rewrites
`NEXT_PUBLIC_API_URL` | No | `https://api.techbes.co.in` | Client-accessible fallback URL for backend APIs
`NEXT_PUBLIC_API_BASE_URL` | No | `https://api.techbes.co.in` | Client-accessible base URL for API requests
`NODE_ENV` | No | `development` / `production` | Controls Next.js compilation mode and caching rules

---

## 13. Build, Setup, and Deployment Guide

### 13.1 Local Development

```
cd c:\Work\technician_app\main_app
npm install
npm run dev
# Main App runs at http://localhost:3000
```

### 13.2 Production Compilation

```
cd c:\Work\technician_app\main_app
npm run build
npm run start
```

---

## 14. Testing Strategy, Verification, and Code Quality Findings

### 14.1 Executed Checks & Empirical Results

1. TypeScript Type-Check (`npx --no-install tsc --noEmit` in `main_app`):
- Result: PASSED (Exit Code: 0).
- Finding: Complete type soundness across all 28 pages, components, and library utilities.

2. Linter Inspection (`npm run lint` in `main_app`):
- Result: FAILED (Exit Code: 1).
- Finding: The package script `"lint": "eslint ."` failed because `eslint` is not installed in `main_app/package.json` dependencies or devDependencies.

3. Automated Unit Tests:
- Result: NO TEST SUITE PRESENT.
- Finding: No test runner (Jest, Vitest) is installed. Two standalone scripts exist: `test-health.js` and `test-sitemap-output.mjs`.

---

## 15. Known Issues, Technical Debt, and Vulnerability Assessment

### 15.1 Confirmed Technical Debt (from Code Evidence)

1. Empty Directory Placeholder: `main_app/app/cctv/[slug]` exists as an empty directory with no `page.tsx`. Dynamic service routing is handled under `/services/[slug]`. This empty directory should be deleted to prevent routing confusion.
2. Missing `eslint` Dependency: `package.json` specifies `"lint": "eslint ."`, but `eslint` is absent from `devDependencies`.
3. Hardcoded Localhost Image Path in `next.config.mjs`: `next.config.mjs` lines 13-14 contain a hardcoded Windows user profile path attempting to copy `hero_illustration_1782651590484.png` from a local directory. This will fail on Linux CI/CD or Vercel production servers if not wrapped in conditional environment guards.
4. Embedded Seeder Inside `next.config.mjs`: `next.config.mjs` lines 28-64 attempt to connect to MongoDB and run `catalogSeeder.js` during Next.js configuration loading. Database seeding logic should be separated from build configuration.

---

## 16. Data Protection, Compliance, and Backup Strategy

1. Customer Address & Geolocation Privacy: Customer coordinates and phone numbers collected during booking are stored in MongoDB Atlas and accessible only by assigned technicians and administrators.
2. Payment Card & UPI Token Safety: No credit card numbers, CVVs, or UPI PINs are handled by TechBes servers. All payment transactions are offloaded directly to Razorpay's PCI-DSS compliant iframe SDK.
3. Compliance Policies: The Main App implements comprehensive statutory legal pages for Indian e-commerce compliance: `/privacy`, `/terms`, `/refund-policy`, `/cancellation-policy`, `/shipping-policy`, `/cookie-policy`, and `/security-policy`.

---

## 17. Operational Troubleshooting and Diagnostics Runbook

Symptom | Diagnostic Root Cause | Verified Remediation Procedure
API calls return HTTP 503 Backend Unavailable | Express backend server (`BE/server.js`) is offline or sleeping on Render. | Start local backend (`cd BE; npm run dev`) or check Render container status. `next.config.mjs` includes an automated wake-up ping.
Customer redirected to `/login` when opening `/dashboard` | `auth-token` cookie is missing or expired. | Normal security behavior. Re-authenticate via `/login` to establish a new 7-day session.
Map pin drop fails to load address name | OpenStreetMap Nominatim rate limit exceeded or network blocked. | Ensure `app/api/geocode/route.ts` sends a compliant User-Agent header. LocationPicker allows manual address text input if geocoding fails.
Voice note upload fails during quotation request | File exceeds 15 MB audio size limit or unsupported format. | Ensure recording format is `.webm`, `.mp3`, `.wav`, or `.m4a` and under 15 MB.
Resident activation returns "Invalid or expired code" | 6-digit code was entered incorrectly or the 48-hour invitation window expired. | Society administrator must open `/admin/apartments` and click Invite again on the resident profile to issue a fresh code.

---

## 18. Handover and Operational Sign-Off Checklist

- [x] Codebase audited in read-only mode without mutations to live production data.
- [x] All 28 customer marketplace modules cataloged with exact file references.

- [x] Reverse proxy rewrites and CORS mitigation verified in `next.config.mjs`.
- [x] Google Gemini AI Assistant integration verified (`AIAssistantWidget.tsx`).
- [x] Multi-step Razorpay checkout flow verified (`cctv-checkout-view.tsx`).
- [x] Society resident activation portal verified (`app/resident/activate/page.tsx`).
- [x] Hyperlocal Bangalore SEO and Answer Engine Optimization datasets cataloged.
- [x] TypeScript compilation verified with 0 errors (`tsc --noEmit`).
- [x] Technical debt in `next.config.mjs` documented for future remediation.

---

## 19. Appendix: Key Source Files Reference

File Path | Lines | Size (Bytes) | Core Architectural Purpose
`main_app/next.config.mjs` | 185 | 5,987 | Reverse proxy rewrites, security headers, redirects, and startup seeder
`main_app/app/amc/page.tsx` | ~1,100 | 47,712 | Interactive AMC pricing calculator and tier comparison
`main_app/components/checkout/cctv-checkout-view.tsx` | ~750 | 30,120 | Checkout wizard with slot picker, saved address, and Razorpay
`main_app/lib/geo-data.ts` | ~1,400 | 60,091 | Hyperlocal Bangalore localities and pin code dataset
`main_app/lib/marketplace-data.ts` | ~680 | 27,984 | Service catalog definitions and fallback marketplace models
`main_app/lib/aeo-data.ts` | ~550 | 23,258 | Answer Engine Optimization semantic search indices
`main_app/components/AIAssistantWidget.tsx` | ~450 | 18,083 | Interactive Google Gemini conversational chatbot widget
`main_app/lib/cctv-api.ts` | 353 | 15,581 | CCTV pricing calculation and hardware catalog models
`main_app/app/dashboard/page.tsx` | ~320 | 12,732 | Customer overview dashboard hub
`main_app/app/services/[slug]/page.tsx` | 142 | 4,436 | Dynamic service page with 301 redirects and catalog lookup
`main_app/app/resident/activate/page.tsx` | 265 | 10,666 | Society flat resident onboarding and account activation
`main_app/proxy.ts` | 25 | 711 | Edge route protection for `/dashboard` and `/login`

*(End of TechBes Main App Technical Documentation)*